



BEST PRACTICE SERIES

Maintenance Planning for OHT Bodies

Application **Maintenance** Site Management Component Rebuild Component Life Management Safety MARC Management

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1.0 Introduction

The maintenance of truck bodies is a vitally important function of the mine maintenance department and, if managed poorly, can have a significant negative impact upon owning & operating costs and machine availability.

This Best Practice describes a process and method for monitoring and trending wear in OHT bodies. Cracks and other body defects are recoded on a plan view of the body and included in a comprehensive report.

The Best Practice incorporates both:

- The condition monitoring procedure and discipline
- The OHT tray/body checklist and data collection matrix.

For various reasons, the bodies at this site had originally been installed with additional armour to control deterioration in high wear areas. When this armour was removed the tray condition deteriorated rapidly over all trucks, increasing workload and making body maintenance unmanageable without a central register.

2.0 Best Practice Description

An Excel database has been developed to capture truck body wear. Today, recording of cracks and other defects is done on a separate template and incorporated into the body condition reports. An Access database version is currently being developed that will include all functions in one and have even greater flexibility.

The database allows the user to select between dual slope and MSD bodies (refer figure 1) and will provide the appropriate inspection point map for the selected body type (refer to Figure 2a)

Figure 1: Header page of Tray Inspection Access Database

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Ultrasonic thickness measurements are then collected against all of these points and entered into the database.

The database allows easy monitoring of the following:

- Tray wear rates for individual bodies (refer Figures 2a & 2b)
- Color coded Tray wear inspection dashboards (hot-sheets) showing wear life remaining in mm and % terms (refer Figure 3)
- Monthly Reporting capability for repairs scheduled and completed (refer Figure 4).

Unit No Tray No

Job Number Date SMU

Service Type Notes:

Loading Board

LB1 LB2 LB3

LB4 LB5 LB6

Left

L1 L2 L3 L4

Right

R1 R2 R3 R4

Floor

F1 F2 F3 F4

F5 F6 F7 F8

F9 F10 F11 F12 F13

Dovetail

D1 D2 D3 D4 D5

Legend:

- L - Left hand inner wall
- LB - Loading Board
- R - Right hand inner
- F - Floor section
- S - 45's beside floor
- D - Dovetail

NB: F11 fwd point of dovetail

Figure 2a: Wear measurement input screen

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Figure 2b: Data from Table 2a is stored in the tray database

Figure 3: Tray wear hot-sheet indicates life remaining in mm and /or % remaining

Completed Scheduled repairs from Tray Inspections			Feb-07
	Repair description	Schedule Date	Complete
1	204, head board repairs.	04/02/2007	Y
2	207, Fit out dovetail centre with wear strips.	06/02/2007	Y
3	214, Plate RH side of floor & strip dove tail.	15/02/2007	Y
4	215, Strip dove tail centre plate.	16/02/2007	Y
5	221, Shovel damage to sides, repair as required.	21/02/2007	Y
6	225, Repair floor rock box.	25/02/2007	Y
7	226, Change back board to head board pressing.	26/02/2007	Y
8	227, Fit new back board plate.	27/02/2007	N
9	230, Replace RH hungry board. Fit wear strips to centre dove tail.	02/03/2007	N
10	224, Tray change.	28/03/2007	Y
11			
12			
13			
14			
15			

Pending Scheduled repairs from Tray Inspections			Feb-07
	Repair description	Schedule Date	Complete
1	205, Inspect condition of RH hungry board.	11/03/2007	
2	209, Fit new hungry board plate.	15/03/2007	
3	224, Repair centre section of dove tail extension.		
4	226, Repair RH side of dove tail extension.		
5			

Create Report

View an existing report

Update contents of existing report

Main Menu

Delete Row

Add Row

Figure 4: Body Maintenance reports can be generated

Importantly, the tool was developed together by WesTrac's on-site Fleet Analyst and Boilermaker Leading Hand. The tool recognizes which of the inspection locations (eg. D2, D3 & D4 in Figure 2a) are on the same liner plate and colour codes the hot-sheets accordingly (eg. Figure 3). This means that the hot-sheets report on the overall condition of a full line plate section rather than an individual point location.

The Tray Inspection dashboard is a very effective tool for tray backlog management. Bodies with locations coloured orange allow the maintenance planner/scheduler to order required materials in advance and schedule labor resources so that when the dashboard turns red for that location, the repairs can be completed with minimized downtime.

For sites that collect the number of cycles completed per truck via VIMS, this can also allow for accurate estimation of body life remaining in terms of time. For example, by knowing the wear rate per 100 cycles (note: a high percentage of the abrasive body wear will occur during the dumping of material on each cycle), then estimating the number of shifts until wear-out should be possible, to a high degree of accuracy, in all inspection locations.

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3.0 Implementation Steps

- Develop the Tray Wear Database to suit your site's needs
- Incorporate Tray Wear Inspections into your site's condition monitoring program with appropriate sign-off process to ensure that inspections always take place. The dealer who developed these check sheets completes the inspections every 250, 500 & 3000 hours.
- The on site Fleet Analyst monitors the wear trends for all bodies
- The planner/schedulers will order required resources as wear trending indicates that sections are reaching the end of their wear life.
- Scheduler updates Backlog Management system
- Project Manager uses database-reporting capabilities to update on the body maintenance program.
- All role descriptions and process maps should be updated to reflect all the above steps

4.0 Benefits

Tony Blake is a Leading Hand boilermaker with WesTrac in Western Australia and is a primary user of this tool.

Benefits realized through this Best Practice include:

- No unscheduled delays due to material shortages
- material is ordered and delivered in advance
- Reduced risk of machine downtime due to unscheduled body repair
- Reduced chassis maintenance costs due to maximizing remaining body life
- Improved scheduling accuracy to enable manpower efficiency
- Use of chassis crack data to better understand crack initiation points and propagation rates
- Improved data collection for maintenance budgeting
- Improved communication between maintenance and production departments relative to chassis management and scheduled chassis repairs

"The tool is good for long term planning. It identifies trays that require major rebuilds. It is also very good for short-term minor in-service repairs, which can be assessed at a glance from the colour coded summary page. We have found it very useful for tracking tray location for cost per hour and repair history. The trays have their own plant number with their own cost centre etc - the program tracks location and SMU a report can be printed"

Tony Blake, Leading Hand Boilermaker
WesTrac – Western Australia

5.0 Resources Required

- Caterpillar part no. 168-7720 Gauge Grp (for ultrasonic thickness testing)
- Excel database as described
- Checklist for use by technicians

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6.0 Supporting Attachments / References

- Caterpillar Mining Bodies Operation and Maintenance Guide (AEDK0627)
- Tray Inspection Excel Database – TrayProgram.xls

7.0 Related Best Practices

0806-2.70-1007	Condition Monitoring - Field Inspections
0806-2.70-1008	Backlog Management

8.0 Acknowledgements

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Rob Ivaldi (Fleet Analyst) & Tony Blake (Boiler-Maker Leading Hand)
 WesTrac Pty Ltd
 Maintenance Department – KCGM Operation
 Western Australia

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